

BACHELOR OF TECHNOLOGY
COMPUTER SCIENCE AND ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY /
INFORMATION TECHNOLOGY / COMPUTER SCIENCE AND TECHNOLOGY
THIRD YEAR (SIXTH SEMESTER)

W.E.F. ADMISSION BATCH 2023-24

Sl. No.	Category	Course Code	Course	Contact Hrs. L-T-P	Credit	University Marks	Internal Evaluation
Subject (Theory)							
1	PC	CSPC3004	Compiler Design	3-0-0	3	100	50
2	PC	CSPC3005	Software Engineering	3-0-0	3	100	50
3	PE	CSPE3008	Evolutionary Computing	3-0-0	3	100	50
		CSPE3009	Data Mining and Data Warehousing				
		CSPE3010	Embedded Systems				
		CSPE3011	Cloud and Edge Computing				
		CSPE3012	Information Security and Management				
		CSPE3013	Software Testing and Quality Assurance				
		CSPE3014	Computer Graphics and Animation				
4	PE	AIPC3005	Deep Learning	3-0-0	3	100	50
		CSPE3016	Big Data Analytics				
		CSPE3017	Advanced Computer Architecture				
		CSPE3018	Internet of Things				
		CSPE3019	Digital Forensics				
		CSPE3020	E-Commerce and ERP				
		CSPE3021	Simulation and Modeling				
5	HS	HSBS3003	Professional Ethics	3-0-0	2	100	50
		HSBS3002	Entrepreneurship Development				
6	MC	MCMC3002	Industrial Safety Engineering	3-0-0	2	100	50
		MCMC3001	Environmental Engineering				
Subject (Sessional / Practical)							
7	PSI	CSPS3202	Project for Product Development - I	0-0-6	3	-	100
8	PC	CSPC3204	Compiler Design Laboratory	0-0-3	1.5	-	100
9	PC	CSPC3205	Software Engineering Laboratory	0-0-3	1.5	-	100
			Total	18-0-12	22	600	600

NB : Minimum 4 weeks of Summer Internship and Research Experience-II (SIRE-II) after 6th Semester during the vacation.

[Click here to view/download the syllabus of the subjects.](#)

CSPC3004 COMPILER DESIGN (3-0-0)

Module I: (08 Hours)

Introduction to Compiler and Lexical Analysis

Language processing systems: preprocessors, compiler, assembler, interpreter, linker, bootstrap loaders, and cross compilers. Structure of a compiler and the different phases.

Lexical Analysis: Role of the lexical analyzer, input buffering, regular expressions, construction of NFA and DFA, DFA minimization, transition diagrams, token recognition. Lexical errors and recovery, introduction to lexical analyzer generator (LEX).

Module II: (08 Hours)

Syntax Analysis

Role of a parser, derivation, ambiguity, left recursion, left factoring, recursive descent parsing.

Bottom-up parsing: Shift-reduce parsing, operator precedence parsing, handles and handle pruning.

Introduction to LR parsing: SLR, CLR, and LALR parsing tables. Parser conflicts and resolution, dangling else problem, error recovery strategies in parsing. Parser generator (YACC).

Module III: (08 Hours)

Semantic Analysis and Syntax-Directed Translation

Semantic Analysis: Syntax-directed definitions (SDD), evaluation of semantic rules.

Translation schemes: Syntax-directed translation schemes (SDTS), implementation of S-attributed and L-attributed definitions. Type checking and conversions. Intermediate Code Generation: Abstract syntax trees, three-address code, quadruples, triples, indirect triples. Translation of expressions, assignment statements, Boolean expressions, case statements, back patching, and procedure calls.

Module IV: (08 Hours)

Symbol Table and Runtime Environment

Symbol table: structure, operations (insert, lookup), scope management, activation records.

Runtime environment: storage organization, stack allocation, heap management, access to non-local data, parameter passing mechanisms. Error handling: classification of errors, error recovery strategies.

Module V: (08 Hours)

Code Optimization and Code Generation

Code optimization:

- Machine-independent: constant folding, common subexpression elimination, dead code elimination, copy propagation, loop optimization, strength reduction, basic blocks and flow graphs, data flow analysis.
- Machine-dependent: DAG-based optimization, peephole optimization, register allocation, instruction scheduling, inter-procedural optimization, garbage collection using reference counting.
- Code generation: target machine model, issues in the design of a code generator, generation of code for expressions and control structures.

CSPC3204 COMPILER DESIGN LABORATORY (0-0-3)

Course Objectives

- To provide hands-on experience in implementing various phases of a compiler.
- To develop skills in using compiler construction tools such as LEX and YACC.
- To enhance understanding of syntax and semantic analysis through practical programming.
- To construct intermediate representations and perform simple code generation.
- To foster practical insights into error handling, symbol table management, and code optimization.

Course Outcomes

Upon successful completion of the lab, students will be able to:

- CO1. Design and implement lexical analyzers using regular expressions and LEX.
- CO2. Develop syntax analyzers using context-free grammars and YACC.
- CO3. Implement semantic analysis and generate intermediate code for simple programming constructs.
- CO4. Create and manage symbol tables and activation records.
- CO5. Understand and apply basic code optimization techniques.
- CO6. Build components of a basic compiler and integrate them for a working solution.

List of Experiments (Indicative)

1. Implementation of Lexical Analyzer using LEX
 - Tokenization of a simple program
 - Handling identifiers, keywords, numbers, and operators
2. Design of Syntax Analyzer using YACC
 - Parsing arithmetic expressions
 - Parsing if-else and while statements
3. Removing Left Recursion and Left Factoring from a Grammar
 - Manual implementation with sample inputs
4. Implementation of Predictive Parser (LL(1))
 - Constructing parsing table and simulating parsing steps
5. Implementation of Shift-Reduce Parser
 - Simulating stack operations for parsing
6. Syntax Directed Translation and Construction of Abstract Syntax Trees
7. Intermediate Code Generation for Expressions and Control Statements
 - Three-address code generation for arithmetic expressions
 - Backpatching for boolean expressions and control flow
8. Design and Implementation of Symbol Table
 - Insertion, lookup, scope handling
9. Implementation of Type Checking and Type Conversion
10. Basic Code Optimization Techniques
 - Constant folding, dead code elimination, strength reduction
11. Simulation of Runtime Environment
 - Activation record structure
 - Stack management for function calls
12. Mini Project: Development of a Simple Compiler for a Subset of a Language
 - Integration of lexical, syntax, and intermediate code generation phases

Software Requirements

- LEX/FLEX
- YACC/Bison
- GCC (C Compiler)
- Linux Environment or Compatible Terminal Emulator (e.g., Cygwin, Ubuntu WSL)

CSPC3005 SOFTWARE ENGINEERING (3-0-0)

Module - I: (06 Hours)

Overview Of Software Engineering

Nature of Software, Software Engineering, Software process, project, product, Process Models, Classical Evolutionary models, Introduction to Agility - Agile Process-Extreme programming - XP Process – Principles of Agile Software Development framework - Overview of System Engineering

Module - II: (08 Hours)

Introduction To Software Project Management

Planning, Scope, Work break-down structure, Milestones, Deliverables, Cost and Estimates - (Human Resources, Time-scale, Costs), Risk Management, RMMM Plan, CASE TOOLS, Agile Project Management, Managing team dynamics and communication, Metrics and Measurement

Module - III: (08 Hours)

Modelling Requirements

Software requirements and its types, Requirements Engineering process, Requirement Elicitation, System Modeling – Requirements Specification and Requirement Validation, Requirements Elicitation techniques, Requirements management in Agile.

Module - IV: (09 Hours)

Software Design

Design concepts and principles - Abstraction - Refinement - Modularity Cohesion coupling, Architectural design, Detailed Design Transaction Transformation, Refactoring of designs, Object oriented Design User-Interface Design

Module - V: (09 Hours)

Validation And Verification

Strategic Approach to Software Testing, Testing Fundamentals Test Plan, Test Design, Test Execution, Reviews, Inspection and Auditing – Regression Testing – Mutation Testing - Object oriented testing - Testing Web based System - Mobile App testing – Mobile test Automation and tools – DevOps Testing – Cloud and Big Data Testing, Software Maintenance, Types of Maintenance, - Software Configuration Management – Overview – SCM Tools. Re-Engineering, Reverse Engineering, Software Reuse

CSPC3205 SOFTWARE ENGINEERING LABORATORY (0-0-3)

Course Objectives:

1. To introduce the essential Software Engineering concepts.
2. To impart concepts and skills for performing analysis, design ,develop, test and evolve efficient software systems of various disciplines and applications
3. To make familiar about engineering practices, standards and metrics for developing software components and products.

Course Outcome:

On completion of this course, student should be able to:

1. Demonstrate the complete Software life cycle activities from requirements analysis to maintenance using the modern tools and techniques.

List of Experiments:

1. Analysis and Identification of the suitable process models
2. Work Break-down Structure (Process Based, Product Based, Geographic Based and Role Based) and Estimations
3. Requirement modelling using Entity Relationship Diagram(Structural Modeling)
4. Requirement modelling using Context flow diagram, DFD (Functional Modeling)
5. Requirement modelling using State Transition Diagram (Behavioral Modeling)
6. OO design – Use case Model, Class Model
7. OO design – Interaction Models
8. OO design – Package, Component and deployment models
9. Design and demonstration of test cases. Functional Testing and Non- Functional Testing (using any open source tools)
10. Story Boarding and User Interface design Modelling

CSPE3011 CLOUD AND EDGE COMPUTING (3-0-0)

Module – I:(08 Hours)

Introduction to Cloud Computing:

Definition and Evolution of Cloud Computing, Cloud Computing Reference Models, Characteristics, Benefits, and Challenges, Service Models (IaaS, PaaS, SaaS), Deployment Models: Public, Private, Hybrid, Community Clouds.

Case Studies: AWS, Google Cloud, Microsoft Azure

Module – II:(08 Hours)

Virtualization and Resource Management:

Virtualization Concepts: VMs, Hypervisors (Type I & II), Containers vs Virtual Machines (Docker, Kubernetes), CPU, Memory, Storage Virtualization, Resource Allocation and Scheduling, Service Level Agreements (SLAs)

Module – III:(08 Hours)

Edge Computing:

Edge Computing Fundamentals, Edge Computing Architecture, Edge vs Cloud , Edge Devices and Gateways, Real-time Processing and Latency Considerations.

From Cloud to Edge computing: Waves of innovation, Edge Computing to support User Applications (5G-Slicing, self-driving cars)

Use Cases: IoT, Smart Cities, Healthcare.

Module – IV:(07 Hours)

Security, Privacy, and Data Management:

Security Threats in Cloud and Edge, Authentication and Access Control, Data Encryption and Protection, Identity and Key Management, GDPR and Regulatory Compliance in Distributed Environments, Secure Data Offloading Techniques.

Module – V:(09 Hours)

Applications, Simulation, and Tools:

Cloud-Edge Integration in Real-World Scenarios, Simulation Tools: iFogSim, CloudSim, EdgeSim, Workload Distribution and Mobility, Performance Evaluation Metrics, Mini Projects: Smart Surveillance, Vehicular Networks, Edge AI, Emerging Trends: Serverless, Green Computing, AI at Edge

CSPE3016 BIG DATA ANALYTICS (3-0-0)

Module - I

Classification of data, Characteristics, Evolution and definition of Big data, What is Big data, Why Big data, Traditional Business Intelligence Vs Big Data, Typical data warehouse and Hadoop environment. Big Data Analytics: What is Big data Analytics, Classification of Analytics, Importance of Big Data Analytics, Technologies used in Big data Environments, Few Top Analytical Tools, NoSQL.

Module - II

Introduction to Hadoop: Introducing Hadoop, Why Hadoop, Why not RDBMS, RDBMS Vs Hadoop, History of Hadoop, Hadoop overview, Use case of Hadoop, HDFS (Hadoop Distributed File System), Processing data with Hadoop, Managing resources and applications with Hadoop YARN(Yet Another Resource Negotiator).

Module - III

Introduction to Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting, Compression. Introduction to MongoDB: What is MongoDB, Why MongoDB, Terms used in RDBMS and MongoDB, Data Types in MongoDB, MongoDB Query Language.

Module - IV

Introduction to Hive: What is Hive, Hive Architecture, Hive data types, Hive file formats, Hive Query Language (HQL), RC File implementation, User Defined Function (UDF). Introduction to Pig: What is Pig, Anatomy of Pig, Pig on Hadoop, Pig Philosophy, Use case for Pig, Pig Latin Overview, Data types in Pig, Running Pig, Execution Modes of Pig, HDFS Commands, Relational Operators, Eval Function, Complex Data Types, Piggy Bank, User Defined Function, Pig Vs Hive.

Module - V

Spark and Big Data Analytics: Spark, Introduction to Data Analysis with Spark. Text, Web Content and Link Analytics: Introduction, Text Mining, Web Mining, Web Content and Web Usage Analytics, Page Rank, Structure of Web and Analyzing a Web Graph.

SHS3002 ENTREPRENEURSHIP DEVELOPMENT (3-0-0)

Module-I: (10 hours)

Entrepreneurship: Concept of entrepreneurship and intrapreneurship, Types of Entrepreneurs, Nature and Importance, Entrepreneurial Traits and Skills, Entrepreneurial Motivation and Achievement, Entrepreneurial Personality

Module-II: (08 hours)

Entrepreneurial Environment, Identification of Opportunities, Converting Business Opportunities into reality. Start-ups and business incubation, Setting up a Small Enterprise. Issues relating to location, Environmental Problems and Environmental pollution Act, Industrial Policies and Regulations

Module-III: (10 hours)

Need to know about Accounting, Working capital Management, Marketing Management, Human Resources Management, and Labour Laws. Organizational support services - Central and State Government, Incentives and Subsidies.

Module-IV: (12 hours)

Sickness of Small-Scale Industries, Causes and symptoms of sickness, cures of sickness, Role of Banks and Governments in reviving industries.

MCMC3001 ENVIRONMENTAL ENGINEERING (3-0-0)

Module-I: (08 Hrs)

Water: Sources of Water and quality issues, water quality requirement for different beneficial uses, Water quality standards, water quality indices, water safety plans, Water Supply systems, Need for planned water supply schemes, Water demand industrial and agricultural water requirements, Components of water supply system; Transmission of water, Distribution system, Various valves used in W/S systems, service reservoirs and design. Water Treatment: aeration, sedimentation, coagulation flocculation, filtration, disinfection, advanced treatments like adsorption, ion exchange, membrane processes.

Module-II: (08 Hrs)

Sewage- Domestic and Storm water, Quantity of Sewage, Sewage flow variations. Conveyance of sewage- Sewers, shapes design parameters, operation and maintenance of sewers, Sewage pumping; Sewerage, Sewer appurtenances, Design of sewerage systems. Small bore systems, Storm Water- Quantification and design of Storm water; Sewage and Sullage, Pollution due to improper disposal of sewage, National River cleaning plans, Wastewater treatment, aerobic and anaerobic treatment systems, suspended and attached growth systems, recycling of sewage – quality requirements for various purposes.

Module-III: (08 Hrs)

Air - Composition and properties of air, Quantification of air pollutants, monitoring of air pollutants, Air pollution- Occupational hazards, Urban air pollution automobile pollution, Chemistry of combustion, Automobile engines, quality of fuel, operating conditions and interrelationship. Air quality standards, Control measures for Air pollution, construction and limitations

Module-IV: (08 Hrs)

Noise-Basic concept, measurement and various control methods. Solid waste Management-Municipal solid waste, Composition and various chemical and physical parameters of MSW, MSW management: Collection, transport, treatment and disposal of MSW. Special MSW: waste from commercial establishments and other urban areas, solid waste from construction activities, biomedical wastes, Effects of solid waste on environment: effects on air, soil, water surface and ground health hazards. Disposal of solid waste-segregation, reduction at source, recovery and recycle. Disposal methods- Integrated solid waste management. Hazardous waste: Types and nature of hazardous waste as per the HW Schedules of regulating authorities.

CSPS3202 PROJECT FOR PRODUCT DEVELOPMENT - I (0-0-6)

Course objectives:

Learning outcomes Upon completing the course, students are expected to be able to:

- Plan and execute independently projects aimed at collecting, systematize and analyze information about markets and customer contexts as fundamental supporting elements for product development in specific sectors and industrial fields,
- Apply the most important models for organizing and managing product development and its implementation in concrete commercial setting,
- Analyze complex product development situations and, based on such analysis, suggest relevant strategies, plans and action programs for various types of companies and organizations,
- Identify the need of further knowledge and tools (both analytics and computer-based) to conduct product development tasks,
- Evaluate critically the result of a product development project and, based on such an evaluation, reflect on uncertainty and risks in execution, so to be able to suggest alternative conclusions.

Instruction

The course is organized as an independent project conducted in teams of 4-5 students with the aim of developing an idea all the way to a product ready for launch for a specific firm. This work is supported by methodological lectures. The project work is presented during a series of seminars with oppositions acting as "control gates" and leading to a final seminar where a decision on launching or not the product is taken by an opponent group.